Generative Project Milestone2

Group 3
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Code repository :
https://github.com/ljmljm7/ie7615-generative-project

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1. Introduction

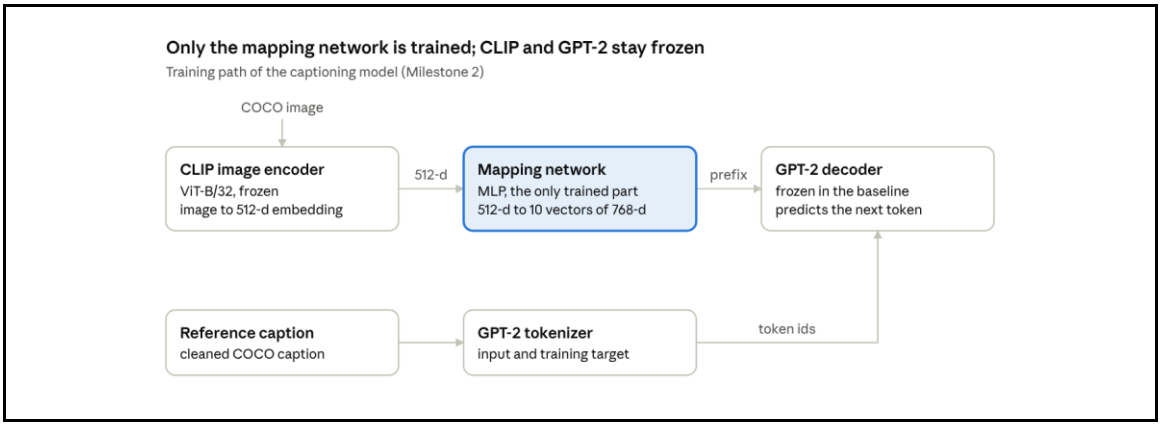
In Milestone 2, we connected the frozen CLIP image embeddings from Milestone 1 to a pretrained GPT-2 decoder through a trained mapping network (prefix conditioning) and trained it on the train split. We then generated captions for 10 test images with three decoding strategies: greedy decoding, beam search (5 beams), and nucleus sampling (top-p = 0.9).

The model reached its best validation loss of **2.674** (perplexity **14.5**) at **epoch 3**. The generated captions usually describe the main subject and scene but miss details such as actions, small objects, and counts. Repetition control removed the repetition loops of greedy decoding and beam search, and beam search with repetition control gave the most fluent captions.

2. Model Integration

We implemented embedding injection by **prefix conditioning** (Figure 1). The L2-normalized CLIP embedding of an image is mapped by a two-layer MLP to 10 vectors of 768 dimensions, the width of GPT-2's token embeddings. These prefix vectors are placed in front of the caption token embeddings and fed to GPT-2 as input embeddings. Only the mapping network is trained; GPT-2 stays frozen (Table 1). The prefix and padding positions are excluded from the loss (label −100), so GPT-2 learns to predict the caption from the first word to the end-of-sequence token.

< Figure 1. Model Architecture: Prefix Conditioning >



< Table 1. Model Components >

Component	Configuration	Parameters	Trained
CLIP image encoder	ViT-B/32, embeddings precomputed in Milestone 1	—	No
Mapping network	Linear 512 → 3,840, Tanh, Dropout 0.1, Linear 3,840 → 7,680, reshaped to 10 × 768	31,468,800	**Yes**
GPT-2 decoder	gpt2 (12 layers, 768 hidden units)	124,439,808	No
Total (without CLIP)		**155,908,608**	20.2% trained

3. Training

Training used the Milestone 1 train split (800 images, 4,000 image–caption pairs), and the validation split (100 images, 500 pairs) tracked the validation loss; the test split was held out for caption generation. Captions were tokenized with the GPT-2 tokenizer, with the end-of-sequence token appended and also used for padding, and padded or truncated to 30 tokens (4 of 5,000 captions were truncated). The configuration is listed in Table 2.

< Table 2. Training Configuration >

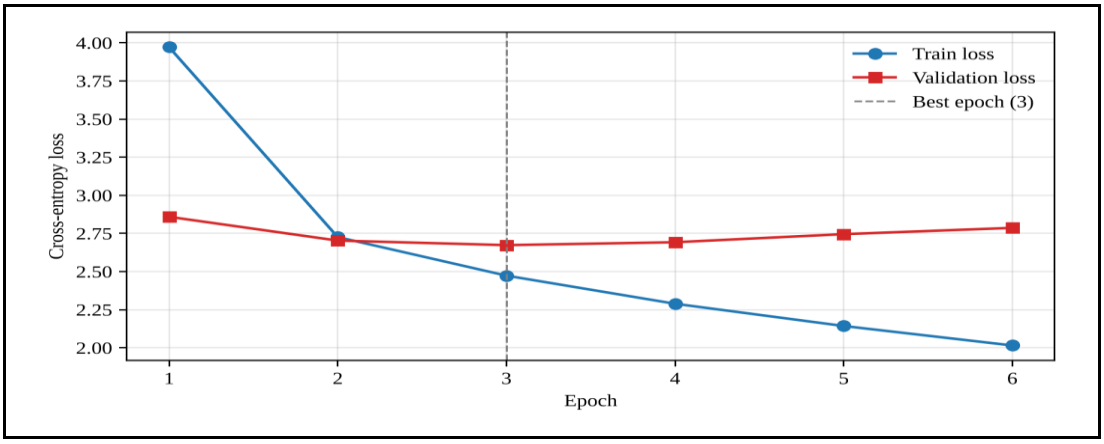
Setting	Value
Optimizer	AdamW, weight decay 0.01
Learning rate	2×10^{-4}, linear warmup for 100 steps, then linear decay
Batch size / epochs	32 (125 steps per epoch) / up to 20, early stopping with patience 3
Loss	Cross-entropy on caption tokens (teacher forcing), gradient clipping 1.0
Hardware	Google Colab, NVIDIA T4 GPU, about 30 s per epoch

Training stopped early after 6 epochs, and the epoch-3 checkpoint with the lowest validation loss was kept (Table 3, Figure 2).

< Table 3. Training Log >

Epoch	Train loss	Val loss	Val perplexity	Learning rate	Time (s)	Checkpoint
1	3.9719	2.8593	17.45	1.98e-04	27.6	Saved
2	2.7263	2.7034	14.93	1.88e-04	28.3	Saved
3	**2.4736**	**2.6736**	**14.49**	**1.77e-04**	**28.9**	**Saved (best)**
4	2.2891	2.6926	14.77	1.67e-04	29.6	
5	2.1433	2.7457	15.58	1.56e-04	30.2	
6	2.0156	2.7868	16.23	1.46e-04	30.8	Early stop

< Figure 2. Training and Validation Loss per Epoch >



4. Caption Generation

Captions were generated from the prefix alone with three decoding strategies on the same trained model (Table 4).

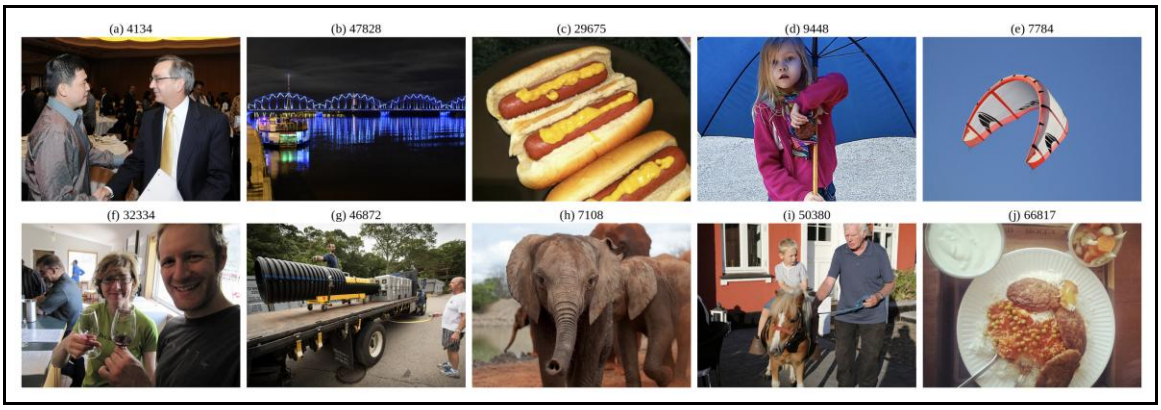
< Table 4. Decoding Strategies >

Strategy	How the next token is chosen
Greedy decoding	The single most probable token at each step. Fast and deterministic, but it cannot revise earlier choices and often produces generic or repetitive captions.
Beam search (5 beams)	Keeps the five most probable partial captions and returns the one with the highest overall score. Deterministic and usually more fluent than greedy, but it can still repeat.
Nucleus sampling (top-p = 0.9)	Samples from the smallest set of tokens whose cumulative probability reaches 90%. More diverse, but captions can drift away from the image.

(a) Initial test : without any constraint, greedy decoding and beam search sometimes fell into repetition loops that ran until the 30-token limit, for example "a man and a woman in a man's man's man's ..." for image 4134. Nucleus sampling was less affected because it samples from the distribution.

(b) Baseline with repetition control : we set *no_repeat_ngram_size* = 3, so any three-token sequence can appear only once in a caption. This removed the long loops, and captions ended naturally at the end-of-sequence token (beam search for 4134: "a man and a woman sitting on a table"). We use this setting as our baseline (Figure 3, Table 5).

< Figure 3. Test Images Used for the Caption Samples >



< Table 5. Baseline Captions on 10 Test Images (no_repeat_ngram_size = 3) >

Image	Reference (1 of 5)	Greedy	Beam search	Nucleus
(a) 4134	a man is shaking hands with another man	a man and a woman in a man's man's chair	a man and a woman sitting on a table	the man and the woman being in a thong
(b) 47828	a blue lighted bridge over water at night	a bridge with a bridge and a train	a wooden bridge with a bridge on top of it	a bridge with a large wooden bridge with some sort of train nearby

Image	Reference (1 of 5)	Greedy	Beam search	Nucleus
(c) 29675	a plate topped with three hot dogs on top of a table	a box of hamburgers is sitting on a table	a box of hamburgers is sitting on a table	a group of three food cabs sitting on a table
(d) 9448	a little girl holds up a big blue umbrella	a young girl standing in the grass	a young girl standing in front of a pool	a woman with her little black scarf on
(e) 7784	this is a large kite flying high in the sky	a man is flying a kite with a kilt	a man is flying a kite in the air	a kite flying on the field
(f) 32334	two people are smiling holding empty wine glasses	a man and woman are standing in a kitchen table	a man holding a glass of wine in his hand	a man is holding a glass of wine and looking out the window
(g) 46872	a man standing along side of a truck trailer	a man is standing on a concrete platform with a hose	a person is standing on a dock with a hose running down the side of the dock	a group of men wearing white boots and a black shirt sit on a concrete deck
(h) 7108	a herd of elephants walking away from a watering hole	a group of elephants standing in a field	a group of elephants standing in front of a fence	a young elephant walking in a field
(i) 50380	a man guiding a pony with a boy riding on it	a man standing on a horse with a man in the background	a man standing on a horse with a man in the background	a man standing on a horse riding in a crowd
(j) 66817	a white plate topped with meat veggies and rice with sauce	a plate of food on a table with a spoon on top	a bowl of food is sitting on a plate on a table	the food is on a plate and is covered in milk

5. Summary of Observations

(a) Training : most of the improvement came in the first epoch, and the validation loss reached its minimum at epoch 3. After that, the training loss kept decreasing (2.47 → 2.02) while the validation loss rose (2.67 → 2.79), which indicates overfitting on the small training set of 800 images; early stopping kept the epoch-3 checkpoint.

(b) Caption content : the model usually captures the main subject and scene (elephants, a kite, a bridge, food on a plate) but misses details: actions (shaking hands in 4134), objects (the umbrella in 9448), and counts (three hot dogs in 29675). Possible causes are that a single 512-dimensional CLIP vector loses fine details, and that 4,000 training pairs may not be enough for the mapping network to pass all image information to GPT-2.

(c) Decoding : repetition control removed the long loops, but short repeats remained ("a bridge with a bridge and a train" in 47828), and a blocked phrase was sometimes replaced with a wrong word ("a man is flying a kite with a kilt" in 7784). Beam search with repetition control gave the most fluent captions, while nucleus sampling was more varied but less faithful to the image.

(d) Frozen GPT-2 : GPT-2 sometimes produced a line break instead of the end-of-sequence token and continued with unrelated web-like text from its pretraining data, so we keep only the first line of each generated caption.

6. Next Steps

Most errors come from missing details rather than from decoding, and the frozen GPT-2 limits how well the model adapts to the image content. In Milestone 3, we will compare three fine-tuning strategies: training only the mapping network (current setting), LoRA on GPT-2, and full fine-tuning of GPT-2. We will also evaluate all three decoding strategies quantitatively with BLEU, METEOR, CIDEr, and ROUGE-L.